

University of San Francisco

Physics 303/CS 486/CS 686 – Deep Learning & Bayesian Learning
Spring, 2026

Homework 1

Due: Tuesday, February 10, 2026, at 11 PM

Machine Learning Metrics and Bayes' Theorem

Brief Introduction

In Machine Learning (ML) and AI, there are a myriad of metrics to assess the performance of training results. Typically, in supervised ML training, the training sample is broken into the training set and the validation set. The training set is fed into the the ML algorithm to optimize the parameters of the ML model — this is called model training. The validation set is “hidden” from this process; it is reserved for measuring the performance of the trained model. In this homework, you will work with two pairs of the most widely used metrics. One pair was defined in class, True Positive Rate (TRP) and False Positive Rate (FPR). The second pair, Precision (p) and Recall (r) are defined as follows:

$$p = \frac{TP}{TP + FP}, \quad r = \frac{TP}{TP + FN} = \frac{TP}{P}$$

A Simple Problems in Two Parts

Suppose your colleague trained a neural network to look for a rare microorganism in microscopic images using a neural network. The expected rate of occurrence of such microorganism is one in 10,000 images.

Your colleague sent you an email that provided the following information:

In the validation set, there are 6300 negative images (without the microorganism) and 190 positive images (with the microorganism). The precision is $p = 0.83$ and the recall is $r = 0.87$.

- a) Find the True Positive Rate (TRP) and False Positive Rate (FPR).
- b) If you now apply this trained neural net to millions of images, on average, out of how many images you inspect do you expect to find one image with the microorganism?

Hint: Out of 10,000 image, how many true positive images do you expect? Out of 10,000 image, how many false positive images do you expect?

Or you can use Bayes' Rule in Week 1 Lecture.